

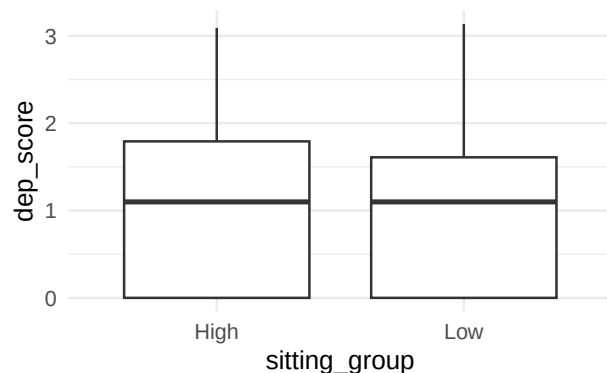
## Day 02: Simple Linear Regression (SLR)

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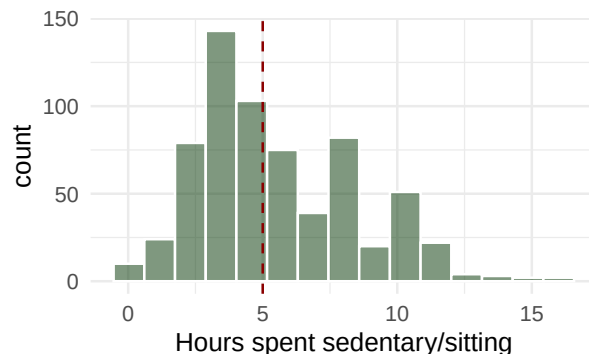
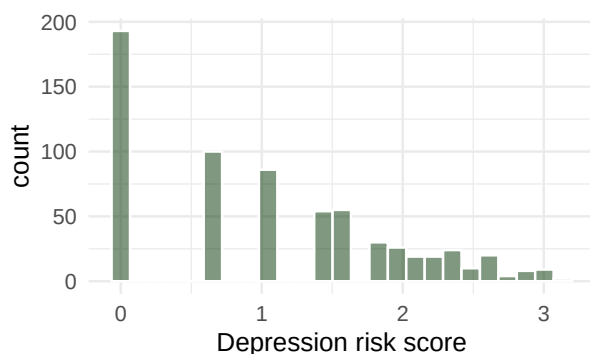
Last time

The *National Health and Nutrition Examination Survey* (NHANES) is an ongoing CDC survey that collects health and behavioral data from a nationally representative sample of the U.S. population. participants aged 18-25 were classified into two groups: those who reported **more** sitting time than average (“High”), and those who reported **less** (“Low”). Their depression symptoms were measured with the PHQ-9, a standard 9-item screening scale (0-27, higher = more symptoms). This was translated into a **depression risk score**, with a larger number indicating a higher risk.

Last time, we used the sample statistics and our trusty intro stat skills to show that there was **no significant difference** between the two groups.



A closer look:



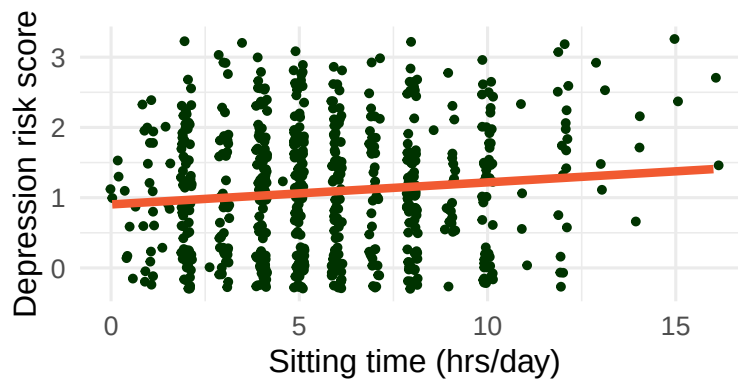
- Response variable: \_\_\_\_\_
- Predictor variable: \_\_\_\_\_

## Regression, Intro Stat version

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The main linear regression skills we focus on in intro stat are:

- Describe the scatterplot
- Write the equation of the regression line
- Interpret the slope and intercept
- Make a prediction



**Example:** describe the scatterplot

There is a \_\_\_\_\_ (strength), \_\_\_\_\_ (direction), \_\_\_\_\_ (form) association between sitting time and depression risk score.

```
lm(dep_score ~ sedentary_hours, data = nhanes)
```

```
(Intercept) sedentary_hours  
0.90426173      0.03140912
```

**Example:** Write out the regression equation

**Example:** Interpreting the slope:

### **i** Correlation vs Causation

The slope describes an *association* – an increase in sitting time doesn't *cause* an increase in depression symptoms just because the model says so. Watch for language that implies causation from an

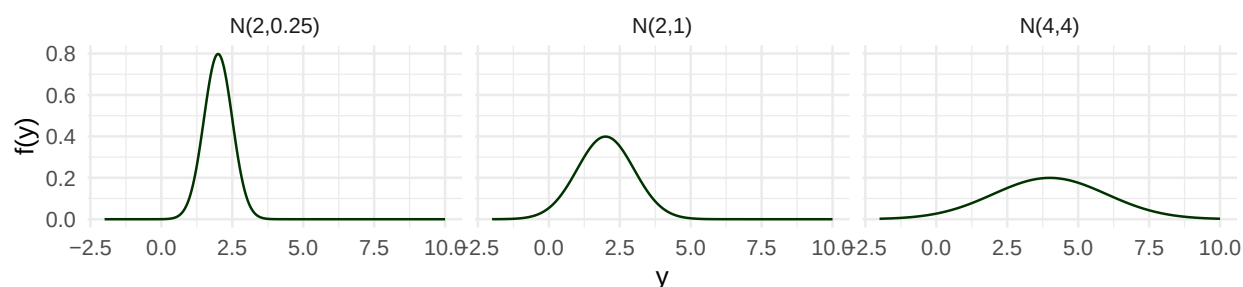
observational study.

**Example:** make a prediction. A participant reports sitting for 6 hours per day. The model predicts:

## Regression, Stat230 Version

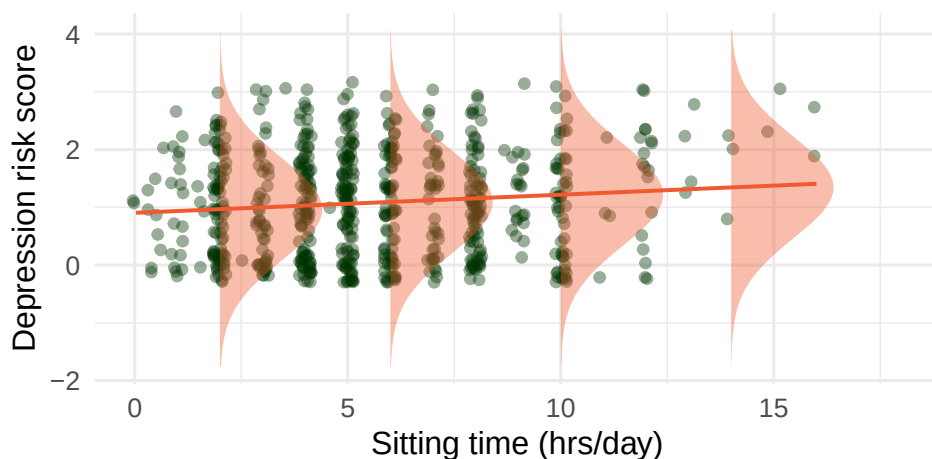
Recap: the Normal distribution

If  $Y$  is **numeric**, **unimodal**, and **symmetric**, we might use a Normal model:  $Y \sim N(\mu, \sigma^2)$ .



The Simple Linear Regression model

$$Y_i = \beta_0 + \beta_1 x_i + \varepsilon_i \quad \text{where} \quad \varepsilon_i \stackrel{\text{iid}}{\sim} N(0, \sigma^2)$$



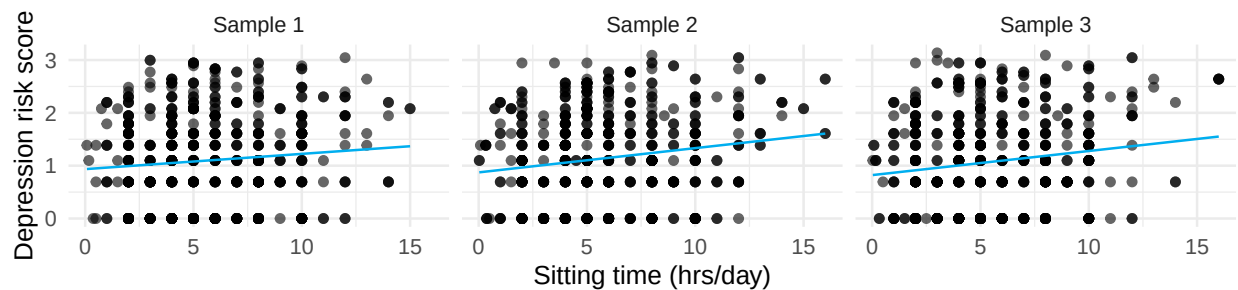
1. Linear relationship between  $x$  and  $y$
2. Errors are independent and identically distributed (iid)
3. Errors are normally distributed
4. Errors have mean 0
5. Variance of the errors doesn't depend on  $x$

## Notation

Mean function:  $E(Y | X) = \mu\{Y | X\} = \beta_0 + \beta_1 x_i$

Fitted model equation:  $\hat{Y}_i = \hat{\mu}\{Y | X\} = \hat{\beta}_0 + \hat{\beta}_1 x_i$

## Sampling Variability (preview for next class!)



## R Activity

Open the SLR in R activity on Moodle and complete it in your groups. Make note of any questions, and choose someone responsible for reporting out your group's answers.

All of the R commands you need are in the file, but here's a reference list of some of the functions you'll learn:

| Function                                             | Purpose                                    |
|------------------------------------------------------|--------------------------------------------|
| <code>read.csv("file_or_url")</code>                 | Load a CSV data set from a file or the web |
| <code>glimpse(data)</code>                           | Get a quick overview of a data set         |
| <code>mean(x)</code>                                 | Calculate the mean                         |
| <code>sd(x)</code>                                   | Calculate the standard deviation           |
| <code>gf_histogram(~x, data = data, bins = n)</code> | Create a histogram                         |
| <code>gf_boxplot(~x, data = data)</code>             | Create a boxplot                           |
| <code>gf_point(y ~ x, data = data)</code>            | Create a scatterplot                       |
| <code>gf_lm()</code>                                 | Add a SLR line to a scatterplot            |
| <code>cor(x, y)</code>                               | Calculate the correlation                  |
| <code>lm(y ~ x, data = data)</code>                  | Fit a SLR model                            |
| <code>summary(model)</code>                          | Get detailed info about a fitted model     |
| <code>confint(model, level = 0.95)</code>            | Confidence interval for SLR coefficients   |
| <code>predict(model, newdata = new_data)</code>      | Make predictions using a fitted model      |